

PAY

TECHNICAL & FUNCTIONAL DOCUMENTATION

Multi-Currency, Multi-Country, Multi-Dimensional Payroll

Field-Level Specifications · CTC-Based Salary Structures
Normal & Statutory Components · Reports · Compliance

Comprehensive technical and functional documentation covering salary structure definition, payroll component master with field-level specifications, normal and statutory component catalogs, multi-currency processing, multi-dimensional cost allocation, payroll input management, input/standard/normal/statutory reports, compliance calendar and filing management, and integration architecture for the global payroll module.

CONFIDENTIAL

Version 1.0 · June 2026

Table of Contents

1. Executive Summary	3
2. System Overview	3
2.1 Multi-Country Architecture	3
2.2 Multi-Currency Framework	3
2.3 Multi-Dimensional Cost Allocation	4
3. Salary Structure Definition (CTC-Based)	4
3.1 CTC Architecture Overview	4
3.2 CTC Template - Field-Level Specification	4
3.3 CTC Component Mapping - Field-Level Specification	5
3.4 CTC Calculation Flow	6
4. Payroll Component Master	6
4.1 Component Master - Field-Level Specification	6
5. Normal Components (Earnings and Deductions)	8
5.1 Normal Earnings Components	8
5.2 Normal Deductions Components	9
6. Statutory Components	10
6.1 Statutory Component Architecture	10
6.2 Statutory Component - Field-Level Specification	10
6.3 Country-Specific Statutory Components	11
7. Income Tax Slab Configuration	14
7.1 Tax Slab Table - Field-Level Specification	14
7.2 Tax Slab Rate Lines - Field-Level Specification	14
8. Payroll Input Management	15
8.1 Input Types and Categories	15
8.2 Payroll Input Record - Field-Level Specification	16
9. Multi-Currency Processing	17
9.1 Currency Configuration - Field-Level Specification	17

9.2 Exchange Rate Table - Field-Level Specification	18
10. Multi-Dimensional Allocation	19
10.1 Dimension Definition - Field-Level Specification	19
10.2 Employee Dimension Allocation - Field-Level Specification	19
11. Payroll Processing Workflow	20
11.1 End-to-End Payroll Processing Stages	20
11.2 Payroll Run Configuration - Field-Level Specification	21
12. Payroll Reports	22
12.1 Input Reports	22
12.2 Standard Reports	23
12.3 Normal Reports	24
12.4 Statutory Reports	25
13. Compliance Management	26
13.1 Compliance Calendar	26
13.2 Compliance Filing Record - Field-Level Specification	27
13.3 Country Compliance Matrix	28
14. Employee Payroll Transaction Record	29
14.1 Payroll Transaction Line - Field-Level Specification	29
15. Payroll Accounting Integration	31
15.1 GL Account Mapping - Field-Level Specification	31
16. Non-Functional Requirements	32
16.1 Performance	32
16.2 Security and Data Privacy	32
16.3 Availability and Scalability	32
16.4 Audit and Traceability	32
17. API Specification Overview	33

1. Executive Summary

This document provides a comprehensive technical and functional specification for the **Multi-Currency, Multi-Country, Multi-Dimensional Payroll Module** within the HRMS platform. The module is engineered to process payroll across multiple legal entities, jurisdictions, currencies, and organizational dimensions simultaneously, ensuring statutory compliance in every operating country while maintaining a single unified configuration and processing framework. The design supports CTC-based salary structure definition, flexible component modeling for both normal (earnings, deductions, allowances) and statutory (tax, social security, provident fund, insurance) components, comprehensive reporting across input, standard, normal, and statutory categories, and automated compliance enforcement with configurable rule engines for each jurisdiction.

The payroll module addresses the complexity inherent in global workforce compensation by abstracting country-specific statutory rules into configurable compliance engines, supporting real-time currency conversion with multi-rate methodologies, enabling multi-dimensional cost allocation across departments, cost centers, projects, locations, and business units, and providing an audit trail that satisfies both internal governance requirements and external regulatory audit standards. The system supports payroll processing for employees across 50+ country jurisdictions, with each country's statutory framework modeled as a pluggable compliance cartridge containing tax brackets, social contribution schedules, reporting templates, and regulatory filing formats.

2. System Overview

2.1 Multi-Country Architecture

The multi-country architecture is built on the principle of a **common payroll engine** with **country-specific compliance cartridges**. The common engine handles universal payroll processing logic: salary structure evaluation, component calculation, currency conversion, dimensional allocation, payroll accounting, and report generation. Each country cartridge encapsulates the jurisdiction-specific rules for income tax computation, social security contributions, statutory deductions, mandatory filings, and regulatory reporting formats. This separation ensures that adding a new country requires developing only the compliance cartridge without modifying the core processing engine. The architecture supports nested legal entities (global parent, regional subsidiary, local branch) with payroll data isolation and consolidated reporting capabilities.

2.2 Multi-Currency Framework

The multi-currency framework enables payroll processing in the employee's local currency while supporting reporting, consolidation, and disbursement in any configured currency. The system maintains exchange rates from an integrated rate provider (or manual entry) with support for multiple rate types: central bank rate, spot rate, monthly average rate, and custom contractual rates. Each payroll component can be designated with a specific currency, and the system automatically computes cross-currency totals, handles currency rounding per locale, and generates accounting entries in both the transaction currency and the base currency of the legal entity. The framework also supports split-currency payroll scenarios where certain components (e.g., international allowance) are paid in a different currency than the base salary.

2.3 Multi-Dimensional Cost Allocation

Multi-dimensional cost allocation distributes payroll costs across organizational dimensions for accurate financial reporting and budget tracking. The supported dimensions include: **Department** (organizational hierarchy), **Cost Center** (financial reporting unit), **Project** (time-bound work allocation), **Location** (geographic office or work site), **Business Unit** (strategic business segment), and **Custom Dimensions** (user-defined classification axes). Each employee's payroll cost can be split across multiple dimension values using percentage-based or amount-based allocation rules. The allocation logic supports effective-dated rules to handle mid-period organizational changes and project reassignments.

3. Salary Structure Definition (CTC-Based)

3.1 CTC Architecture Overview

The Cost to Company (CTC) model defines the total expenditure an organization incurs for an employee, encompassing all direct compensation (base pay, allowances, bonuses), employer-side statutory contributions (provident fund, social security, insurance), and non-cash benefits (company car, accommodation). The salary structure module allows HR administrators to define CTC templates that decompose the total CTC into individual components with configurable allocation percentages, fixed amounts, or formula-driven calculations. Each CTC template is country-specific, accounting for local statutory requirements and common compensation practices, while remaining flexible enough to accommodate organization-specific customization.

3.2 CTC Template - Field-Level Specification

Field Name	Data Type	Length	Required	Description
ctc_template_id	VARC HAR	32	Yes	Unique identifier for the CTC template (auto-generated UUID)
ctc_template_name	VARC HAR	100	Yes	Descriptive name of the CTC template (e.g., "India Standard CTC", "US Salaried CTC")
country_code	CHAR	3	Yes	ISO 3166-1 alpha-3 country code for which this template is applicable
legal_entity_id	VARC HAR	32	No	Specific legal entity if template is entity-specific; NULL = applicable to all entities in country
currency_code	CHAR	3	Yes	ISO 4217 currency code for the CTC amount (default currency for the country)
ctc_type	ENUM	-	Yes	ANNUAL / MONTHLY - Defines whether the CTC amount represents annual or monthly cost
effective_from	DATE	-	Yes	Start date from which this template version is active
effective_to	DATE	-	No	End date; NULL = currently active, set when superseded by new version
version	INT	-	Yes	Template version number for audit tracking, incremented on each modification

Field Name	Data Type	Length	Required	Description
status	ENUM	-	Yes	DRAFT / ACTIVE / INACTIVE - Lifecycle status of the template
is_default	BOOLEAN	-	Yes	TRUE if this is the default template for the country; only one default per country per entity
base_pay_pct	DECIMAL	5,2	No	Default percentage of CTC allocated to base pay (e.g., 40.00 for 40%)
created_by	VARCHAR	64	Yes	User ID of the template creator
created_date	TIMESTAMP	-	Yes	Timestamp of template creation (auto-populated)
modified_by	VARCHAR	64	No	User ID of last modifier
modified_date	TIMESTAMP	-	No	Timestamp of last modification

Table 1: CTC Template - Field Definitions

3.3 CTC Component Mapping - Field-Level Specification

Each CTC template contains a set of component mappings that define how the total CTC is decomposed into individual payroll components. The component mapping specifies the allocation method (percentage of CTC, fixed amount, formula), the component type (earning, deduction, employer contribution), and the processing order for sequential calculation.

Field Name	Data Type	Length	Required	Description
mapping_id	VARCHAR	32	Yes	Unique identifier for the component mapping record
ctc_template_id	VARCHAR	32	Yes	Foreign key referencing the parent CTC template
component_id	VARCHAR	32	Yes	Foreign key to the payroll component master definition
component_name	VARCHAR	80	Yes	Display name of the component (e.g., "Basic Salary", "HRA", "Employer PF")
component_category	ENUM	-	Yes	EARNING / DEDUCTION / EMPLOYER_CONTRIBUTION / REIMBURSEMENT / BENEFIT_IN_KIND
allocation_method	ENUM	-	Yes	PERCENTAGE_OF_CTC / FIXED_AMOUNT / PERCENTAGE_OF_COMPONENT / FORMULA / PRORATA
allocation_value	DECIMAL	12,2	Yes *	Percentage value (if method = PERCENTAGE_OF_CTC) or fixed amount (if FIXED_AMOUNT). *Required when allocation_method is PERCENTAGE or FIXED

Field Name	Data Type	Length	Required	Description
base_component_id	VARCHAR	32	No	Reference component ID when allocation_method = PERCENTAGE_OF_COMPONENT (e.g., HRA as 40% of Basic)
formula_expression	TEXT	-	No	Formula expression when allocation_method = FORMULA (e.g., "BASIC * 0.12 + MIN(1800, CTC*0.03)")
calculation_sequence	INT	-	Yes	Order in which components are calculated (1 = first). Essential for dependent calculations
is_statutory	BOOLEAN	-	Yes	TRUE if this is a statutory component (PF, ESI, Tax, etc.) that cannot be removed by user
is_taxable	BOOLEAN	-	Yes	TRUE if the component is subject to income tax
tax_exemption_limit	DECIMAL	12,2	No	Maximum amount exempt from taxation under applicable tax law (e.g., HRA exemption limit)
frequency	ENUM	-	Yes	MONTHLY / QUARTERLY / ANNUAL / ONCE / VARIABLE - Payment frequency of the component
proration_rule	ENUM	-	No	CALENDAR_DAYS / WORKING_DAYS / FIXED_MONTH - How to prorate for mid-period joins/leaves
min_amount	DECIMAL	12,2	No	Minimum value the component can take (enforcement floor)
max_amount	DECIMAL	12,2	No	Maximum value the component can take (enforcement ceiling)
rounding_rule	ENUM	-	No	NEAREST / UP / DOWN / NEAREST_10 / NEAREST_100 - Rounding logic applied after calculation

Table 2: CTC Component Mapping - Field Definitions

3.4 CTC Calculation Flow

The CTC-based salary calculation follows a sequential process: (1) The total CTC amount is entered for the employee; (2) Components are calculated in their defined **calculation_sequence** order; (3) Percentage-based components are computed against the CTC (or their base component); (4) Formula-based components are evaluated using the formula engine with previously computed component values available as variables; (5) Employer-side statutory contributions are computed based on the applicable country cartridge rules; (6) A validation step ensures that the sum of all components (earnings + employer contributions + benefits in kind) equals the total CTC, with the residual automatically allocated to an "Adjustment" component if a rounding discrepancy exists; (7) The resulting monthly breakdown is stored as the employee's active salary structure with effective dating for historical tracking.

4. Payroll Component Master

4.1 Component Master - Field-Level Specification

The Payroll Component Master is the central registry of all earnings, deductions, allowances, reimbursements, and statutory components available in the system. Each component is defined once in the master and can be referenced by multiple CTC templates, salary structures, and country cartridges. The master captures the component's identity, classification, calculation logic, tax treatment, and applicability rules at a granular field level.

Field Name	Data Type	Length	Required	Description
component_id	VARCHAR	32	Yes	Unique component identifier (auto-generated)
component_code	VARCHAR	20	Yes	Short code for the component (e.g., "BASIC", "HRA", "EPF_EE", "PIT") - must be unique per country
component_name	VARCHAR	100	Yes	Full display name of the component
component_type	ENUM	-	Yes	EARNING / DEDUCTION / EMPLOYER_CONTRIB / EMPLOYEE_CONTRIB / REIMBURSEMENT / BENEFIT_IN_KIND / INFORMATIONAL
component_category	ENUM	-	Yes	NORMAL / STATUTORY - Classifies the component as a normal organizational component or a statutory/government-mandated component
country_code	CHAR	3	Yes	Country for which this component is applicable (ISO 3166-1 alpha-3)
calculation_type	ENUM	-	Yes	FLAT_AMOUNT / PERCENTAGE / SLAB_BASED / FORMULA / LOOKUP_TABLE / MANUAL_ENTRY
default_value	DECIMAL	14,2	No	Default flat amount or percentage value used when no employee-specific override exists
percentage_base	VARCHAR	32	No	Component ID of the base on which percentage is calculated (e.g., BASIC for PF @ 12% of Basic)
formula_id	VARCHAR	32	No	Reference to the formula definition in the formula engine when calculation_type = FORMULA
slab_table_id	VARCHAR	32	No	Reference to the slab/rate table when calculation_type = SLAB_BASED (e.g., income tax slabs)
is_taxable	BOOLEAN	-	Yes	Whether the component is subject to income tax withholding
tax_treatment	ENUM	-	No	FULLY_TAXABLE / PARTIALLY_TAXABLE / EXEMPT / DEFERRED - Detailed tax treatment classification
exemption_section	VARCHAR	20	No	Tax section under which exemption applies (e.g., "80C", "80D" for India; "401k" for US)
max_exemption_amount	DECIMAL	14,2	No	Maximum exemption amount per tax year under the applicable section
affects_gross	BOOLEAN	-	Yes	TRUE if this component is included in gross salary calculation

Field Name	Data Type	Length	Required	Description
affects_net	BOOLEAN	-	Yes	TRUE if this component affects net pay (most components do, except informational ones)
affects_ctc	BOOLEAN	-	Yes	TRUE if this component contributes to the Cost to Company calculation
payment_frequency	ENUM	-	Yes	MONTHLY / QUARTERLY / SEMI_ANNUAL / ANNUAL / ONE_TIME / VARIABLE
proration_applicable	BOOLEAN	-	Yes	Whether this component is prorated for partial periods (new join, resignation, LOA)
rounding_rule	ENUM	-	No	NEAREST / CEILING / FLOOR / NEAREST_10 / NEAREST_100 / BANKERS_ROUNDING
is_active	BOOLEAN	-	Yes	Active status flag; inactive components cannot be assigned to new salary structures
effective_from	DATE	-	Yes	Date from which this component definition is effective
effective_to	DATE	-	No	Date until which this component definition is effective (NULL = currently active)

Table 3: Payroll Component Master - Field Definitions

5. Normal Components (Earnings and Deductions)

5.1 Normal Earnings Components

Normal earnings components represent the compensation elements paid to an employee that are not mandated by statute but are defined by the organization's compensation policy. These include fixed components (basic salary, housing allowance, transport allowance), variable components (performance bonus, commission, overtime), and benefit components (meal allowance, communication allowance, education assistance). Each normal earning has configurable calculation rules, tax treatment, and CTC inclusion settings.

Component	Code	Typical Calc.	Taxable	CTC Included	Common Countries	Description
Basic Salary	BASIC	% of CTC (30-50%)	Yes	Yes	All	Foundation earning; basis for most statutory calculations
House Rent Allowance	HRA	% of Basic (40-50%)	Partially	Yes	India, Pakistan	Housing allowance with tax exemption up to defined limits
Transport Allowance	TRA	Fixed / % Basic	Partially	Yes	India, UAE	Commuting cost reimbursement with partial or full exemption
Special Allowance	SPA	Residual CTC balancer	Yes	Yes	India	Catch-all component to balance CTC after other allocations

Component	Code	Typical Calc.	Taxable	CTC Included	Common Countries	Description
Dearness Allowance	DA	% of Basic (CPI-linked)	Yes	Yes	India, Bangladesh	Cost of living adjustment linked to consumer price index
Medical Allowance	MED	Fixed monthly	Partially	Yes	India, Pakistan	Fixed medical reimbursement; partially exempt with bills
Leave Travel Allowance	LTA	Fixed / Annual	Partially	Yes	India	Tax-exempt travel allowance for domestic trips (2-year block)
Overtime Pay	OT	Hourly rate x hours	Yes	No	All	Compensation for hours worked beyond standard work week
Performance Bonus	PERF B	Variable / Annual	Yes	Yes	All	Discretionary or formulaic bonus based on performance rating
Commission	COM M	% of sales / deals	Yes	No	All	Sales-driven variable pay calculated on revenue targets
Shift Allowance	SHIFT	Fixed per shift	Yes	Yes	All	Additional pay for night/weekend/holiday shift work
Hardship Allowance	HARD	Fixed / % Basic	Partially	Yes	Multiple	Additional pay for difficult or remote work locations
International Allowance	INTL	Fixed (foreign currency)	Varies	Yes	Global	Allowance for expatriates paid in assignment-country currency

Table 4: Normal Earnings Components Catalog

5.2 Normal Deductions Components

Normal deductions are non-statutory withholdings from employee pay as defined by organizational policy. These include loan repayments, insurance premiums (voluntary), salary advances, and voluntary savings contributions. Unlike statutory deductions, normal deductions can be configured, added, or removed by HR administrators based on employee elections and company policy.

Component	Code	Calculation	Pre-Tax/Post-Tax	Frequency	Description
Salary Advance Recovery	ADV_REC	Fixed / Installment	Post-Tax	Monthly	Recovery of advance paid to employee, typically in 3-6 installments
Employee Loan EMI	LOAN_EMI	Fixed installment	Post-Tax	Monthly	Equated monthly installment for company-provided loans
Voluntary PF	VPF	% of Basic (above statutory)	Pre-Tax	Monthly	Employee-elected additional PF contribution beyond statutory minimum
Voluntary Insurance	VINS	Fixed premium	Post-Tax	Monthly	Premium for supplemental health/life insurance elected by employee
Professional Membership	PMEM	Fixed / Annual	Post-Tax	Monthly/Annual	Deduction for professional body membership fees

Component	Code	Calculation	Pre-Tax/ Post-Tax	Frequen cy	Description
Meal Card Deduction	MEAL_D	Fixed monthly	Pre-Tax	Monthly	Deduction for subsidized meal card programs (e.g., Sodexo)
Union Dues	UNION	Fixed / % Basic	Post-Tax	Monthly	Labor union membership dues as per collective agreement
Garnishment	GARN	Court-ordered amount	Post-Tax	Monthly	Court-mandated deduction for debt repayment, child support, etc.
Parking Deduction	PARK	Fixed monthly	Post-Tax	Monthly	Employee contribution for company parking facility
Recovery (Overpayment)	REC_OV	Variable	Post-Tax	One-time/ Monthly	Recovery of previously overpaid amounts identified in audit

Table 5: Normal Deductions Components Catalog

6. Statutory Components

6.1 Statutory Component Architecture

Statutory components are government-mandated deductions and contributions that must be calculated, withheld, and remitted in strict compliance with local laws. The system models each country's statutory framework as a **compliance cartridge** containing the applicable taxes, social security schemes, and insurance programs. Each statutory component is linked to a calculation engine that implements the exact legal formula, including slab rates, contribution ceilings, age-based variations, and filing status adjustments. The statutory component definition includes remittance scheduling, filing requirements, and penalty calculation rules to support end-to-end compliance management.

6.2 Statutory Component - Field-Level Specification

Field Name	Data Type	Length	Required	Description
statutory_component_id	VARCHAR	32	Yes	Unique identifier for the statutory component
component_code	VARCHAR	20	Yes	Short code (e.g., "EPF_EE", "EPF_ER", "ESI_EE", "PIT", "SOC_SEC")
component_name	VARCHAR	100	Yes	Full statutory name (e.g., "Employees Provident Fund - Employee Share")
country_code	CHAR	3	Yes	ISO country code for jurisdiction
authority_name	VARCHAR	150	Yes	Name of the government authority (e.g., "EPFO", "HMRC", "IRS", "CPF Board")
authority_code	VARCHAR	30	No	Official code for the authority (used in electronic filing)

Field Name	Data Type	Length	Required	Description
party_type	ENUM	-	Yes	EMPLOYEE / EMPLOYER / BOTH - Who makes the contribution or payment
calculation_basis	ENUM	-	Yes	GROSS_PAY / BASIC_PAY / SPECIFIC_COMPONENT / SLAB_BASED / FLAT_RATE
basis_component_id	VARCHAR	32	No	Component ID of the basis component when calculation_basis = SPECIFIC_COMPONENT
rate_percentage	DECIMAL	6,3	Yes *	Contribution rate as percentage (*when applicable; slab-based uses slab_table_id)
wage_ceiling	DECIMAL	14,2	No	Maximum amount of basis pay subject to this contribution (e.g., EPF ceiling INR 15,000/month)
max_contribution_amt	DECIMAL	14,2	No	Absolute maximum contribution amount per period regardless of basis
min_contribution_amt	DECIMAL	14,2	No	Minimum mandatory contribution (floor amount)
slab_table_id	VARCHAR	32	No	Reference to progressive slab/rate table for income tax type calculations
remittance_frequency	ENUM	-	Yes	MONTHLY / QUARTERLY / ANNUAL / ON_DEMAND - How often amounts must be remitted to authority
remittance_due_day	INT	-	No	Day of month by which remittance must be received (e.g., 15th for EPF in India)
filing_frequency	ENUM	-	No	MONTHLY / QUARTERLY / ANNUAL / HALF_YEARLY - Frequency of statutory return filing
filing_format	VARCHAR	30	No	Format code for electronic filing (e.g., "ECR", "FPS", "W-2", "IR8A")
penalty_rate_pct	DECIMAL	5,2	No	Penalty interest rate per month for late remittance (e.g., 1.00 for 1%/month)
is_challan_required	BOOLEAN	-	Yes	Whether a payment challan must be generated before remittance
challan_format	VARCHAR	30	No	Challan format code (e.g., "CHALLAN_ITNS", "PAYE302")
effective_from	DATE	-	Yes	Date from which this statutory rule is effective
effective_to	DATE	-	No	End date; NULL = currently in force

Table 6: Statutory Component - Field Definitions

6.3 Country-Specific Statutory Components

6.3.1 India Statutory Components

Component	Code	Party	Rate	Basis	Ceiling	Frequency	Authority
EPF - Employee	EPF_EE	Employee	12%	Basic + DA	INR 15,000/mo	Monthly	EPFO
EPF - Employer	EPF_ER	Employer	12% (3.67% PF + 8.33% EPS)	Basic + DA	INR 15,000/mo	Monthly	EPFO
EPS (Pension)	EPS_ER	Employer	8.33%	Basic + DA	INR 15,000/mo	Monthly	EPFO
EDLI - Employer	EDLI_ER	Employer	0.5%	Basic + DA	INR 15,000/mo	Monthly	EPFO
EDLI Admin Charges	EDLI_ADM	Employer	0.01%	Basic + DA	INR 15,000/mo	Monthly	EPFO
ESI - Employee	ESI_EE	Employee	0.75%	Gross Salary	INR 21,000/mo	Monthly	ESIC
ESI - Employer	ESI_ER	Employer	3.25%	Gross Salary	INR 21,000/mo	Monthly	ESIC
Professional Tax	PTAX	Employee	Slab-based	Gross Salary	Varies by state	Monthly	State Govt
TDS (Income Tax)	TDS	Employee	Slab-based	Taxable Income	None	Monthly	Income Tax Dept
Labour Welfare Fund	LWF_EE	Employee	Fixed / Slab	State-specific	State-specific	Half-yearly	State Govt
LWF - Employer	LWF_ER	Employer	Fixed / Slab	State-specific	State-specific	Half-yearly	State Govt

Table 7: India - Statutory Components

6.3.2 United States Statutory Components

Component	Code	Party	Rate	Basis	Ceiling	Frequency	Authority
Federal Income Tax	FIT	Employee	Slab (7 brackets)	Taxable Wages	None	Per pay period	IRS
State Income Tax	SIT	Employee	Varies by state	State Taxable Wages	Varies	Per pay period	State Revenue
Social Security (OASDI)	SS_EE	Employee	6.2%	Gross Wages	\$168,600/yr (2024)	Per pay period	SSA
Social Security (OASDI)	SS_ER	Employer	6.2%	Gross Wages	\$168,600/yr	Per pay period	SSA
Medicare (HI)	MED_EE	Employee	1.45% + 0.9% Surtax	All Wages	None (\$200k surtax)	Per pay period	IRS
Medicare (HI)	MED_ER	Employer	1.45%	All Wages	None	Per pay period	IRS

Component	Code	Party	Rate	Basis	Ceiling	Frequency	Authority
FUTA	FUTA	Employer	6.0% (0.6% effective)	First \$7,000 wages	\$7,000/yr	Quarterly	IRS
SUTA	SUTA	Employer	Varies by state	State wage base	Varies	Quarterly	State Agency
SDI (CA, NJ, etc.)	SDI	Employee	Varies	State wage base	Varies	Per pay period	State Agency

Table 8: United States - Statutory Components

6.3.3 United Kingdom Statutory Components

Component	Code	Party	Rate	Basis	Ceiling	Frequency	Authority
Income Tax (PAYE)	PAYE	Employee	Slab (20/40/45%)	Taxable Pay	None	Per pay period	HMRC
National Insurance - Employee	NI_EE	Employee	8% (Primary)	Earnings above PT	UEL applies	Per pay period	HMRC
National Insurance - Employer	NI_ER	Employer	13.8% (Secondary)	Earnings above ST	None	Per pay period	HMRC
Student Loan Deduction	SLD	Employee	9% / 6%	Income above threshold	None	Per pay period	HMRC
Workplace Pension (AE)	PEN_EE	Employee	5% (min)	Qualifying Earnings	LEL/UEL band	Per pay period	TPR/Pension Provider
Workplace Pension (AE)	PEN_ER	Employer	3% (min)	Qualifying Earnings	LEL/UEL band	Per pay period	TPR/Pension Provider
Apprenticeship Levy	APP_LEVY	Employer	0.5%	Total Paybill	GBP 15,000 allowance	Monthly	HMRC

Table 9: United Kingdom - Statutory Components

6.3.4 Singapore Statutory Components

Component	Code	Party	Rate	Basis	Ceiling	Frequency	Authority
Central Provident Fund - EE	CPF_EE	Employee	5-20% (age-based)	Ordinary Wages	OW Ceiling SGD 6,800/mo	Monthly	CPF Board
Central Provident Fund - ER	CPF_ER	Employer	7.5-17% (age-based)	Ordinary Wages	OW Ceiling SGD 6,800/mo	Monthly	CPF Board
SDL	SDL	Employer	0.25%	Total Remuneration	None (min SGD 2/mo)	Monthly	SkillsFuture SG

Component	Code	Party	Rate	Basis	Ceiling	Frequency	Authority
FWL	FWL	Employer	Fixed (sector-based)	Per foreign worker	Varies by sector/tier	Monthly	MOM
Income Tax (IRAS)	IRAS	Employee	Slab (0-22%)	Annual Taxable Income	None	Annual / Monthly via AIS	IRAS
Ethnic Fund (CDAC/MBMF/etc)	EFUND	Employee	Fixed by income slab	Monthly Wage	Income slab ceiling	Monthly	Respective Council

Table 10: Singapore - Statutory Components

7. Income Tax Slab Configuration

7.1 Tax Slab Table - Field-Level Specification

Income tax in most jurisdictions is calculated using progressive slab rates where different portions of taxable income are taxed at increasing rates. The tax slab table defines these rate bands with effective dating to handle annual budget changes. The system supports multiple slab tables per country to accommodate different filing statuses (single, married, head of household), different tax regimes (old vs. new in India), and age-based variations (senior citizen rates).

Field Name	Data Type	Length	Required	Description
slab_table_id	VARCHAR	32	Yes	Unique identifier for the slab table
slab_table_name	VARCHAR	100	Yes	Descriptive name (e.g., "India New Regime FY 2024-25", "US Single Filer 2024")
country_code	CHAR	3	Yes	Country for which these slabs apply
tax_year	VARCHAR	10	Yes	Tax year / financial year (e.g., "2024-25", "2024")
filing_status	ENUM	-	No	SINGLE / MARRIED_JOINT / MARRIED_SEPARATE / HEAD_OF_HOUSEHOLD / SENIOR_CITIZEN
regime_type	ENUM	-	No	OLD / NEW / STANDARD - Applicable for countries with multiple tax regime options
effective_from	DATE	-	Yes	Start date of applicability
effective_to	DATE	-	No	End date; NULL = currently applicable

Table 11: Tax Slab Table - Field Definitions

7.2 Tax Slab Rate Lines - Field-Level Specification

Field Name	Data Type	Length	Required	Description
slab_line_id	VARCHAR	32	Yes	Unique identifier for the slab line
slab_table_id	VARCHAR	32	Yes	Foreign key to parent slab table
sequence	INT	-	Yes	Order of the slab (1 = lowest income band)
income_from	DECIMAL	14,2	Yes	Lower bound of income for this slab (inclusive)
income_to	DECIMAL	14,2	No	Upper bound of income (NULL = no upper limit, i.e., highest slab)
rate_percentage	DECIMAL	6,3	Yes	Tax rate percentage applicable for this slab
fixed_amount	DECIMAL	14,2	No	Fixed tax amount added for this slab (used in cumulative calculation method)
surcharge_rate	DECIMAL	5,2	No	Additional surcharge percentage applicable above certain income thresholds
cess_rate	DECIMAL	5,2	No	Health/Education cess percentage applied on total tax (e.g., 4% for India)

Table 12: Tax Slab Rate Lines - Field Definitions

8. Payroll Input Management

8.1 Input Types and Categories

Payroll inputs are the variable data elements required for each payroll period that cannot be derived from the employee's static salary structure alone. The input management module provides a structured framework for collecting, validating, and processing these variable inputs before payroll calculation. Inputs are classified into categories based on their source, frequency, and validation requirements. The system supports bulk upload (CSV/Excel), API-based integration (from attendance, leave, and expense systems), and manual entry through the payroll administrator interface.

Input Category	Examples	Source System	Input Frequency	Validation Rules
Attendance Inputs	Days worked, absent days, half days, work from home days	Attendance / Time Management	Monthly / Bi-weekly	Days worked + Leaves = Total calendar days; No negative values
Leave Inputs	Paid leave days, sick leave, casual leave, loss of pay days	Leave Management	Monthly	Leave balance >= consumed; No overlap with attendance

Input Category	Examples	Source System	Input Frequency	Validation Rules
Overtime Inputs	OT hours (regular, double, holiday), comp-off earned	Time & Attendance	Monthly / Bi-weekly	OT hours <= max allowed per month; Approval status = Approved
Variable Pay Inputs	Bonus amount, commission, incentive, variable pay %	Performance / Sales	Monthly / Quarterly / Annual	Amount within configured min-max range; Manager approval required
One-Time Payments	Joining bonus, retention bonus, ex-gratia, severance	HR / Management	As needed	Approval workflow completed; Proration rule applied if needed
Reimbursements	Medical bills, travel claim, fuel, phone, internet	Expense Management	Monthly	Claim amount <= annual limit; Bills attached; Manager approved
Loan/Advance	New loan amount, advance requested, EMI changes	HR / Finance	As needed	Outstanding balance check; Max loans limit; EMI <= % of net pay
Investment Declarations	80C, 80D, HRA exemption, rent paid, home loan interest	Self-Service Portal	Annual / Quarterly update	Section-wise limit validation; Proof submission deadline check
Tax Adjustments	Additional tax, reduced tax, foreign tax credit, prior period adjustment	Payroll Admin	As needed	Adjustment reason mandatory; Impact on annual projection shown
Organizational Changes	Transfer, promotion, salary revision, designation change	HR Core	As needed	Effective date within payroll period; Approval chain completed

Table 13: Payroll Input Categories and Validation

8.2 Payroll Input Record - Field-Level Specification

Field Name	Data Type	Length	Required	Description
input_id	VARCHAR	32	Yes	Unique identifier for the input record
payroll_run_id	VARCHAR	32	Yes	Foreign key to the payroll processing run for which this input applies
employee_id	VARCHAR	32	Yes	Employee for whom this input is recorded
input_type	ENUM	-	Yes	ATTENDANCE / LEAVE / OVERTIME / VARIABLE_PAY / ONE_TIME / REIMBURSEMENT / LOAN / INVESTMENT / TAX_ADJ / ORG_CHANGE
component_code	VARCHAR	20	Yes	Payroll component code this input maps to (e.g., "OT", "PERFB", "MED_CLAIM")

Field Name	Data Type	Length	Required	Description
input_value_numeric	DECIMAL	14,2	No	Numeric input value (amount or quantity depending on unit_type)
unit_type	ENUM	-	No	AMOUNT / DAYS / HOURS / PERCENTAGE - Unit of measurement for the input value
input_date_from	DATE	-	No	Start date of the input period (e.g., OT from date)
input_date_to	DATE	-	No	End date of the input period
currency_code	CHAR	3	No	Currency of monetary inputs (defaults to employee's payroll currency)
exchange_rate	DECIMAL	12,6	No	Exchange rate applied if input currency differs from payroll currency
approval_status	ENUM	-	Yes	PENDING / APPROVED / REJECTED - Workflow status of the input
approved_by	VARCHAR	64	No	User ID of the approver
approval_date	TIMESTAMP	-	No	Timestamp of approval action
source	ENUM	-	Yes	MANUAL / BULK_UPLOAD / API_INTEGRATION / SELF_SERVICE / SYSTEM_GENERATED
reference_document	VARCHAR	200	No	Reference to supporting document (expense claim ID, leave request ID, etc.)
remarks	TEXT	-	No	Free-text notes or justification for the input
created_by	VARCHAR	64	Yes	User or system that created the input record
created_date	TIMESTAMP	-	Yes	Creation timestamp

Table 14: Payroll Input Record - Field Definitions

9. Multi-Currency Processing

9.1 Currency Configuration - Field-Level Specification

Field Name	Data Type	Length	Required	Description
currency_config_id	VARCHAR	32	Yes	Unique identifier

Field Name	Data Type	Length	Required	Description
legal_entity_id	VARCHAR	32	Yes	Legal entity to which this configuration applies
base_currency	CHAR	3	Yes	ISO 4217 base currency of the legal entity (e.g., "USD", "INR", "GBP")
payroll_currency	CHAR	3	Yes	Currency in which payroll is processed for the entity
reporting_currency	CHAR	3	No	Currency for consolidated group reporting (if different from base)
exchange_rate_source	ENUM	-	Yes	CENTRAL_BANK / REUTERS / BLOOMBERG / MANUAL / CUSTOM_API - Source of exchange rates
rate_type	ENUM	-	Yes	SPOT / MONTHLY_AVERAGE / CUSTOM / CONTRACTUAL - Which rate to use for conversion
auto_fetch_enabled	BOOLEAN	-	Yes	Whether to automatically fetch rates from the configured source
fetch_frequency	ENUM	-	No	DAILY / WEEKLY / MONTHLY / ON_DEMAND - How often rates are refreshed
rounding_precision	INT	-	Yes	Decimal places for rounding after conversion (typically 2)
rounding_rule	ENUM	-	Yes	NEAREST / UP / DOWN / BANKERS - Rounding method for converted amounts
gain_loss_account	VARCHAR	32	No	GL account code for posting exchange rate gains/losses

Table 15: Currency Configuration - Field Definitions

9.2 Exchange Rate Table - Field-Level Specification

Field Name	Data Type	Length	Required	Description
rate_id	VARCHAR	32	Yes	Unique identifier
from_currency	CHAR	3	Yes	Source currency ISO code
to_currency	CHAR	3	Yes	Target currency ISO code
exchange_rate	DECIMAL	14,6	Yes	Conversion rate: 1 from_currency = rate to_currency
rate_date	DATE	-	Yes	Date for which this rate is applicable
rate_type	ENUM	-	Yes	SPOT / MONTHLY_AVERAGE / QUARTERLY_AVERAGE / CUSTOM / CONTRACTUAL
source	VARCHAR	50	Yes	Source of the rate (e.g., "RBI", "ECB", "Manual Entry")
inverse_rate	DECIMAL	14,6	No	Auto-computed inverse rate for reverse conversion

Field Name	Data Type	Length	Required	Description
is_active	BOOLEAN	-	Yes	Active flag; only one active rate per from-to-type-date combination

Table 16: Exchange Rate Table - Field Definitions

10. Multi-Dimensional Allocation

10.1 Dimension Definition - Field-Level Specification

Field Name	Data Type	Length	Required	Description
dimension_id	VARCHAR	32	Yes	Unique identifier for the dimension
dimension_code	VARCHAR	20	Yes	Short code (e.g., "DEPT", "CC", "PROJ", "LOC", "BU")
dimension_name	VARCHAR	80	Yes	Display name (e.g., "Department", "Cost Center", "Project")
dimension_type	ENUM	-	Yes	STANDARD / CUSTOM - Standard dimensions are system-defined; custom are user-defined
hierarchy_enabled	BOOLEAN	-	Yes	Whether the dimension supports parent-child hierarchy for drill-down reporting
allocation_method	ENUM	-	Yes	PERCENTAGE / AMOUNT / HEADCOUNT / FTE - Default method for splitting costs
is_mandatory	BOOLEAN	-	Yes	Whether every employee must have at least one value for this dimension
allow_multiple	BOOLEAN	-	Yes	Whether an employee can be allocated to multiple values of this dimension
max_allocations	INT	-	No	Maximum number of dimension values an employee can be allocated to
effective_from	DATE	-	Yes	Start date of dimension availability

Table 17: Dimension Definition - Field Definitions

10.2 Employee Dimension Allocation - Field-Level Specification

Field Name	Data Type	Length	Required	Description
allocation_id	VARCHAR	32	Yes	Unique identifier

Field Name	Data Type	Length	Required	Description
employee_id	VARCHAR	32	Yes	Employee whose cost is being allocated
dimension_id	VARCHAR	32	Yes	Foreign key to dimension definition
dimension_value_id	VARCHAR	32	Yes	Specific dimension value (e.g., Department = "Engineering", Cost Center = "CC-1001")
allocation_pct	DECIMAL	5,2	Yes	Percentage of total cost allocated to this dimension value (must sum to 100% per dimension)
allocation_amount	DECIMAL	14,2	No	Fixed amount allocation (used when allocation_method = AMOUNT)
effective_from	DATE	-	Yes	Start date of this allocation
effective_to	DATE	-	No	End date; NULL = currently active
is_primary	BOOLEAN	-	Yes	TRUE for the primary/default dimension value; used when single-value reporting is needed

Table 18: Employee Dimension Allocation - Field Definitions

11. Payroll Processing Workflow

11.1 End-to-End Payroll Processing Stages

The payroll processing workflow follows a structured sequence of stages, each with defined inputs, processing steps, validation gates, and outputs. The workflow is designed to ensure data integrity, statutory compliance, and audit traceability at every stage. Payroll administrators can configure the workflow to include or skip optional stages based on organizational requirements.

Stage	Actions	Validation Gate	Output	SLA
1. Payroll Period Open	Create payroll run; Lock master data changes; Initialize period parameters	Period not already open; No pending prior-period runs	Payroll run record with status OPEN	Day 1 of period
2. Input Collection	Collect attendance, leave, OT, variable pay, reimbursements; Import from integrated systems	All mandatory inputs received; Approval workflows completed; No duplicate entries	Validated input dataset ready for processing	Day 1-5
3. Input Verification	HR/admin reviews inputs; Exception reports generated; Manual corrections applied	Exception count below threshold; All critical inputs verified	Verified and approved input dataset	Day 5-7

Stage	Actions	Validation Gate	Output	SLA
4. Payroll Calculation	Execute salary structure evaluation; Calculate earnings, deductions, statutory components; Apply tax projections	Calculation engine completes without errors; All components computed; Currency conversions applied	Pre-processing payroll results (draft)	Day 7-8
5. Compliance Validation	Verify statutory calculations; Check tax projections; Validate against regulatory rules; Generate compliance alerts	All statutory components within tolerance; No compliance violations; Filing deadlines verified	Compliance-validated payroll results	Day 8-9
6. Review and Approval	Payroll manager reviews draft results; Annotates exceptions; Approves or returns for correction	Manager approval obtained; Variance from prior period within tolerance	Approved payroll results	Day 9-10
7. Payroll Accounting	Generate journal entries; Post to GL; Create cost center allocations; Process currency conversions	Accounting entries balanced (debit = credit); Cost allocations sum to 100%	Accounting entries and GL postings	Day 10-11
8. Payslip Generation	Generate employee payslips; Publish to self-service portal; Email notifications sent	Payslips generated for all active employees; Format matches country template	Published payslips	Day 11-12
9. Bank File Generation	Create bank payment files; Generate wire transfer instructions; Process split deposits	Bank file format validated; Total amounts match net pay; Bank account verification passed	Bank payment files ready for upload	Day 12-13
10. Statutory Filing	Generate challans, returns, and statutory reports; Prepare electronic filing data; Submit to authorities	Filing data matches payroll totals; Format validated against authority specification	Filed statutory returns and challans	Per statutory due dates
11. Payroll Close	Lock payroll period; Archive records; Update YTD balances; Reset period inputs	All disbursements confirmed; All filings completed; No pending corrections	Closed payroll period	Day 15+

Table 19: End-to-End Payroll Processing Workflow

11.2 Payroll Run Configuration - Field-Level Specification

Field Name	Data Type	Length	Required	Description
payroll_run_id	VARCHAR	32	Yes	Unique identifier for the payroll run
legal_entity_id	VARCHAR	32	Yes	Legal entity for which payroll is being processed
payroll_period	VARCHAR	10	Yes	Period identifier (e.g., "2026-06" for June 2026)

Field Name	Data Type	Length	Required	Description
period_start_date	DATE	-	Yes	First day of the payroll period
period_end_date	DATE	-	Yes	Last day of the payroll period
pay_date	DATE	-	Yes	Date on which net pay is disbursed to employees
run_type	ENUM	-	Yes	REGULAR / OFF_CYCLE / SUPPLEMENTAL / FINAL_SETTLEMENT / CORRECTION
run_status	ENUM	-	Yes	OPEN / INPUT_COLLECTION / PROCESSING / REVIEW / APPROVED / ACCOUNTING / DISBURSED / CLOSED
currency_code	CHAR	3	Yes	Processing currency for this run
exchange_rate_date	DATE	-	No	Date used for exchange rate determination in this payroll period
tax_projection_method	ENUM	-	Yes	CUMULATIVE / NON_CUMULATIVE / ANNUALIZED - Method for projecting annual tax liability
include_statutory	BOOLEAN	-	Yes	Whether statutory components are calculated in this run
processing_mode	ENUM	-	Yes	FULL / DELTA - Full re-calculation or incremental changes only
initiated_by	VARCHAR	64	Yes	User who initiated the payroll run
initiated_date	TIMESTAMP	-	Yes	Timestamp of run initiation

Table 20: Payroll Run Configuration - Field Definitions

12. Payroll Reports

12.1 Input Reports

Input reports provide visibility into the data collected for payroll processing, enabling administrators to verify completeness, identify missing inputs, and resolve discrepancies before calculation begins. These reports are critical for payroll accuracy as errors in input data propagate through all subsequent calculations and are significantly more expensive to correct after payroll closure.

Report Name	Code	Purpose	Key Columns	Filters	Format
Attendance Summary Report	RPT_ATT_SUM	Verify attendance data before payroll run	Employee, Present Days, Absent Days, Leaves, Holidays, Total Days	Department, Location, Period	PDF / Excel

Report Name	Code	Purpose	Key Columns	Filters	Form at
Missing Inputs Report	RPT_MISS_INP	Identify employees with missing or incomplete payroll inputs	Employee, Missing Input Type, Component, Due Status, Approver	Input Type, Department, Period	PDF / Excel
Input Variance Report	RPT_INP_VAR	Compare current period inputs against previous period to detect anomalies	Employee, Component, Previous Value, Current Value, Variance %, Flag	Department, Variance Threshold, Period	PDF / Excel
Overtime Summary Report	RPT_OT_SUM	Review overtime hours and cost before processing	Employee, OT Type, Hours, Rate, Amount, Approval Status	Department, OT Type, Period	PDF / Excel
Leave Encashment Report	RPT_LEAVE_ENC	List of leave encashments to be processed in current period	Employee, Leave Type, Days Encashed, Rate, Amount, Balance Remaining	Leave Type, Department, Period	PDF / Excel
Investment Declaration Report	RPT_INVEST_DECL	Employee tax-saving declarations for TDS calculation	Employee, Section, Declared Amount, Proof Submitted, Verified Amount	Section, Proof Status, Period	PDF / Excel

Table 21: Input Reports Catalog

12.2 Standard Reports

Standard reports are the core operational reports generated as part of every payroll cycle. These reports provide the primary financial and administrative outputs of payroll processing and are used by payroll teams, finance departments, and management for disbursement, accounting, and decision-making. Standard reports are auto-generated upon payroll approval and are available for download from the payroll run dashboard.

Report Name	Code	Purpose	Key Columns	Filters	Format
Payroll Register	RPT_PAY_REG	Complete payroll calculation detail for all employees in the period	Employee, Component, Amount, Gross, Deductions, Net Pay, YTD	Department, Location, Cost Center, Period, Currency	PDF / Excel
Payroll Summary Report	RPT_PAY_SUM	Aggregated payroll cost by component across the organization	Component, Headcount, Total Amount, Average, Prior Period, Variance %	Department, Entity, Country, Period, Currency	PDF / Excel
Department-Wise Payroll	RPT_DEPT_PAY	Payroll cost breakdown by department with headcount	Department, Headcount, Gross, Total Deductions, Net, Employer Contributions	Entity, Location, Period	PDF / Excel
Cost Center Report	RPT_COST_PAY	Payroll cost allocation by cost center with dimensional split	Cost Center, Allocated Amount, Percentage, Currency, GL Code	Department, Entity, Period, Currency	PDF / Excel

Report Name	Code	Purpose	Key Columns	Filters	Format
Bank Disbursement Report	RPT_B ANK_D ISB	Payment file summary for bank upload and reconciliation	Employee, Bank Account, Amount, Currency, Payment Reference	Bank, Currency, Period	PDF / Excel / Bank File
Payslip Register	RPT_P AYSLIP	Bulk payslip generation and delivery confirmation	Employee, Net Pay, Delivery Status, Delivery Method	Department, Period, Delivery Status	PDF (per employee)
Salary Register (CTC)	RPT_C TC_RE G	CTC breakdown for all employees with component-wise split	Employee, CTC, Basic, Allowances, Statutory ER, Benefits, Monthly Gross	Department, Grade, Entity, Period	PDF / Excel
GL Posting Report	RPT_G L_POS T	Journal entries generated from payroll for GL integration	GL Account, Debit, Credit, Cost Center, Currency, Reference	Entity, Period, GL Range, Currency	PDF / Excel / CSV
Currency Conversion Report	RPT_C URR_C ONV	Details of all cross-currency conversions applied in payroll	Employee, Component, Original Amount, Original Currency, Rate, Converted Amount	Currency Pair, Entity, Period	PDF / Excel
New Joiner Payroll Report	RPT_N EW_JOI N	Payroll details for employees joining in current period	Employee, Join Date, Proration Days, Gross, Net, Bank Details	Department, Period	PDF / Excel
Exit/Final Settlement Report	RPT_FI NAL_S ET	Full and final settlement calculations for separated employees	Employee, Last Day, Earned, Deductions, Leave Encashment, Recovery, Net Payout	Department, Separation Type, Period	PDF / Excel

Table 22: Standard Reports Catalog

12.3 Normal Reports

Normal reports provide ongoing operational insights into payroll trends, employee compensation analytics, and organizational cost patterns. Unlike standard reports which are tied to a specific payroll run, normal reports typically span multiple periods and support trend analysis, benchmarking, and workforce cost planning. These reports are available on-demand and can be scheduled for periodic generation.

Report Name	Code	Purpose	Key Columns	Filters	Format
Month-over-Month Payroll Trend	RPT_MO M_TREN D	Track payroll cost trends across months	Month, Headcount, Gross, Net, Statutory, Total Cost, MoM Variance %	Department, Entity, Year, Currency	PDF / Excel
Salary Structure Distribution	RPT_SA L_DIST	Analyze salary distribution across bands/grades	Salary Band, Headcount, % of Workforce, Avg CTC, Min, Max	Department, Grade, Entity	PDF / Excel

Report Name	Code	Purpose	Key Columns	Filters	Format
Component-wise Cost Analysis	RPT_CMP_COST	Detailed breakdown of payroll cost by component over time	Component, Period, Amount, % of Gross, Trend Direction	Department, Component Type, Period Range	PDF / Excel
Employee Pay History	RPT_EMP_HIST	Historical payroll details for individual employees	Employee, Period, Gross, Net, Components (all), Changes	Employee, Period Range	PDF / Excel
Headcount Cost Report	RPT_HC_COST	Average cost per employee by dimension	Dimension, Headcount, Total Cost, Avg Cost, Median, Cost/FTE	Department, Location, Period, Currency	PDF / Excel
Proration Report	RPT_PRORATE	Details of mid-period joins, exits, and salary changes	Employee, Event, Event Date, Prorated Days, Full Amount, Prorated Amount	Event Type, Department, Period	PDF / Excel
Loan and Advance Report	RPT_LOAN_ADV	Outstanding loan balances and EMI tracking	Employee, Loan Type, Principal, EMI, Outstanding, Installments Remaining	Loan Type, Department, Status	PDF / Excel
Overtime Cost Analysis	RPT_OT_ANALYSIS	Overtime cost trends and patterns by department	Department, OT Hours, OT Cost, % of Gross, Period, Trend	Department, Period Range	PDF / Excel

Table 23: Normal Reports Catalog

12.4 Statutory Reports

Statutory reports are mandatory reports required by government authorities for tax withholding, social security contributions, and regulatory compliance. These reports must be generated in the exact format specified by the relevant authority, with many jurisdictions now requiring electronic filing in prescribed formats (XML, CSV, EDI). The system auto-generates statutory reports upon payroll closure and maintains a filing calendar to track submission deadlines.

Report Name	Code	Country	Frequency	Authority	Filing Format	Due Date
PF ECR (Electronic Challan cum Return)	IND_PF_ECR	India	Monthly	EPFO	CSV (ECR format)	15th of following month
ESI Half-Yearly Return	IND_ESI_RET	India	Half-yearly	ESIC	Online portal	12th Jan / 12th Jul
TDS Return (Form 24Q)	IND_TDS_24Q	India	Quarterly	Income Tax Dept	NSDL format (TXT)	Last day of following quarter month
Form 16 (TDS Certificate)	IND_FORM16	India	Annual	Income Tax Dept	PDF (prescribed format)	15th June

Report Name	Code	Country	Frequency	Authority	Filing Format	Due Date
Professional Tax Return	IND_PT AX_RE T	India	Monthly / Annual	State Govt	State-specific format	Varies by state
W-2 Wage Statement	US_W2	US	Annual	IRS / SSA	EFW2 (SSA format)	Jan 31
941 Quarterly Return	US_941	US	Quarterly	IRS	IRS e-file	Last day of month following quarter
1095-C (ACA Reporting)	US_1095C	US	Annual	IRS	AIR format	March 2 (paper) / March 31 (e-file)
P60 (End of Year Certificate)	UK_P60	UK	Annual	HMRC	PDF (prescribed)	May 31
P11D (Benefits in Kind)	UK_P11D	UK	Annual	HMRC	Online / PDF	July 6
FPS (Full Payment Submission)	UK_FPS	UK	Per pay period	HMRC	RTI XML	On or before pay date
IR8A (Tax Return)	SG_IR8A	Singapore	Annual	IRAS	AIS (Auto-Inclusion Scheme)	March 1
CPF Contribution Report	SG_CPF_RET	Singapore	Monthly	CPF Board	CPF EZPay format	14th of following month
SDL/FWL Return	SG_SDL_FWL	Singapore	Monthly	SkillsFuture / MOM	CPF EZPay format	14th of following month

Table 24: Statutory Reports Catalog (by Country)

13. Compliance Management

13.1 Compliance Calendar

The compliance calendar tracks all statutory filing obligations across all operating countries, providing a unified view of upcoming deadlines, completed filings, and overdue obligations. Each compliance obligation is defined with its frequency, due date calculation rule, responsible party, and escalation chain. The system sends automated reminders at configurable intervals before the due date and escalates to senior management when filings are at risk of missing deadlines. The compliance calendar integrates with the payroll workflow to ensure that filings cannot be prepared until the corresponding payroll run is closed and approved.

Field Name	Data Type	Length	Required	Description
compliance_id	VARCHAR	32	Yes	Unique identifier for the compliance obligation
compliance_name	VARCHAR	150	Yes	Name of the compliance filing (e.g., "Monthly PF ECR Filing")
country_code	CHAR	3	Yes	Country of the compliance obligation
authority_name	VARCHAR	150	Yes	Regulatory authority to which the filing is submitted
filing_type	ENUM	-	Yes	RETURN / CHALLAN / CERTIFICATE / REGISTRATION / REPORT / DECLARATION
frequency	ENUM	-	Yes	MONTHLY / QUARTERLY / HALF_YEARLY / ANNUAL / ONE_TIME
due_date_rule	VARCHAR	50	Yes	Calculation rule for due date (e.g., "D+15" = 15th of following month, "Q_END+30")
grace_days	INT	-	No	Additional grace days beyond the calculated due date before penalty applies
penalty_type	ENUM	-	No	FIXED_AMOUNT / PERCENTAGE_PER_DAY / PERCENTAGE_PER_MONTH / PROGRESSIVE
penalty_value	DECIMAL	10,2	No	Penalty amount or rate (interpreted based on penalty_type)
responsible_role	VARCHAR	50	Yes	Role responsible for completing the filing (e.g., "PAYROLL_ADMIN", "HR_MANAGER")
escalation_role	VARCHAR	50	No	Role to escalate to if filing is not completed by due date
reminder_days_before	VARCHAR	20	No	Comma-separated days before due date for reminder alerts (e.g., "7,3,1")
auto_generate	BOOLEAN	-	Yes	Whether the system automatically generates the filing upon payroll closure
filing_format	VARCHAR	30	No	Format of the filing output (e.g., "ECR_CSV", "EFW2", "RTI_XML")
is_active	BOOLEAN	-	Yes	Active status of this compliance obligation

Table 25: Compliance Calendar - Field Definitions

13.2 Compliance Filing Record - Field-Level Specification

Field Name	Data Type	Length	Required	Description
filing_id	VARCHAR	32	Yes	Unique identifier for the filing record

Field Name	Data Type	Length	Required	Description
compliance_id	VARCHAR	32	Yes	Foreign key to compliance obligation definition
payroll_run_id	VARCHAR	32	Yes	Payroll run from which this filing was generated
filing_period	VARCHAR	10	Yes	Period for which the filing applies (e.g., "2026-06")
filing_status	ENUM	-	Yes	GENERATED / UNDER_REVIEW / SUBMITTED / ACKNOWLEDGED / REJECTED / PENALTY_APPLIED
generated_date	TIMESTAMP	-	Yes	Date the filing was generated by the system
submitted_date	TIMESTAMP	-	No	Date the filing was submitted to the authority
acknowledgement_ref	VARCHAR	50	No	Reference number/acknowledgement from the authority upon acceptance
filing_amount	DECIMAL	14,2	No	Total amount associated with this filing (contribution + penalty if any)
penalty_amount	DECIMAL	14,2	No	Penalty amount applied (if late filing)
file_path	VARCHAR	500	No	Path to the generated filing file (CSV, XML, PDF)
submitted_by	VARCHAR	64	No	User who submitted the filing
reviewed_by	VARCHAR	64	No	User who reviewed the filing before submission
rejection_reason	TEXT	-	No	Reason for rejection if authority rejected the filing
resubmission_date	TIMESTAMP	-	No	Date of resubmission after rejection

Table 26: Compliance Filing Record - Field Definitions

13.3 Country Compliance Matrix

The following matrix summarizes the key compliance obligations by country, highlighting the regulatory framework, key filing requirements, and penalty structures. This matrix serves as a reference for understanding the scope of statutory compliance management required in each jurisdiction.

Country	Key Regulations	Major Filings	Penalty for Late Filing	Electronic Filing
India	EPF Act 1952, ESI Act 1948, IT Act 1961, PT Act (state)	PF ECR, ESI Return, TDS 24Q, Form 16, PT Return, LWF Return	12% p.a. interest (PF), 18% p.a. (ESI), 1-2% per month (TDS)	Mandatory for PF, ESI, TDS
United States	FICA, FUTA, IRC, ACA, State tax codes	941, W-2, 1095-C, SUTA, State returns	2-15% penalty on underpayment; \$50/W-2 late	Mandatory for W-2, 941, 1095-C
United Kingdom	PAYE Regs 2003, NI Act, Pensions Act 2008, Finance Act	FPS, EPS, P60, P11D, PSA, Apprenticeship Levy	1-5% of tax due per month; daily penalties for RTI late	Mandatory RTI (FPS/EPS)
Singapore	CPF Act, Income Tax Act, Employment Act, SDL Act	IR8A, IR8S, CPF return, SDL/FWL return	Up to SGD 5,000 + 6 months jail; CPF 18% p.a. interest	Mandatory via AIS, EZPay
UAE	Labour Law, DIFC/ADGM Employment Regulations, GST	EOSB filing, WPS submission, VAT return (if applicable)	Fines AED 5,000-50,000; WPS non-compliance penalties	Mandatory WPS; Online portals
Australia	SG Act, ITAA 1936/1997, Fair Work Act, STP regime	STP reporting, Payment Summary, Super guarantee, BAS	SG charge + interest + admin; ATO penalties	Mandatory STP Phase 2

Table 27: Country Compliance Matrix

14. Employee Payroll Transaction Record

14.1 Payroll Transaction Line - Field-Level Specification

The payroll transaction line captures the computed value of each payroll component for an employee in a specific payroll run. This is the most granular level of payroll data and serves as the foundation for all reports, payslips, accounting entries, and statutory filings. Every component calculated for an employee generates a transaction line with full audit traceability.

Field Name	Data Type	Length	Required	Description
transaction_id	VARCHAR	32	Yes	Unique identifier for the transaction line
payroll_run_id	VARCHAR	32	Yes	Foreign key to the payroll run
employee_id	VARCHAR	32	Yes	Employee for whom this transaction is recorded
component_id	VARCHAR	32	Yes	Foreign key to the payroll component master

Field Name	Data Type	Length	Required	Description
component_code	VARCHAR	20	Yes	Component short code (denormalized for query performance)
component_type	ENUM	-	Yes	EARNING / DEDUCTION / EMPLOYER_CONTRIB / EMPLOYEE_CONTRIB / REIMBURSEMENT / INFO
component_category	ENUM	-	Yes	NORMAL / STATUTORY
calculated_amount	DECIMAL	14,2	Yes	Amount computed by the payroll calculation engine before any manual override
override_amount	DECIMAL	14,2	No	Manually overridden amount (if a manual adjustment was applied)
final_amount	DECIMAL	14,2	Yes	Actual amount used = override_amount if not NULL, else calculated_amount
currency_code	CHAR	3	Yes	Transaction currency
exchange_rate	DECIMAL	12,6	No	Rate used if currency conversion was applied
base_currency_amount	DECIMAL	14,2	No	Amount converted to legal entity base currency
ytd_amount	DECIMAL	14,2	Yes	Year-to-date cumulative amount for this component and employee
mtd_amount	DECIMAL	14,2	Yes	Month-to-date cumulative amount (relevant for semi-monthly payroll)
proration_factor	DECIMAL	5,4	No	Proration factor applied (1.0000 = full period; e.g., 0.5000 = half period)
input_source_id	VARCHAR	32	No	Reference to the input record that triggered this calculation (if input-driven)
formula_trace	TEXT	-	No	Calculation trace showing formula, variable values, and step-by-step computation for audit
dimension_split_json	JSON	-	No	Multi-dimensional allocation split (e.g., {"DEPT": {"ENG": 0.6, "SALES": 0.4}})
gl_account_code	VARCHAR	30	No	GL account code to which this transaction line posts
cost_center_code	VARCHAR	30	No	Cost center for dimensional accounting
is_reversal	BOOLEAN	-	Yes	TRUE if this is a reversal of a previous period transaction
reversed_transaction_id	VARCHAR	32	No	Reference to the original transaction being reversed
tax_treatment	ENUM	-	No	FULLY_TAXABLE / PARTIALLY_TAXABLE / EXEMPT / PRE_TAX / POST_TAX

Field Name	Data Type	Length	Required	Description
statutory_reference	VARCHAR	50	No	Reference to statutory rule applied (e.g., "EPF_12PCT_15000CEIL")

Table 28: Payroll Transaction Line - Field Definitions

15. Payroll Accounting Integration

15.1 GL Account Mapping - Field-Level Specification

The GL account mapping defines how each payroll component is posted to the organization's general ledger. The mapping supports multi-dimensional accounting with separate debit and credit accounts for each component, cost center, and organizational unit. The system generates balanced journal entries from the approved payroll results and posts them to the integrated ERP/accounting system via API or file-based exchange.

Field Name	Data Type	Length	Required	Description
mapping_id	VARCHAR	32	Yes	Unique identifier
legal_entity_id	VARCHAR	32	Yes	Legal entity for this mapping
component_id	VARCHAR	32	Yes	Payroll component being mapped
debit_account	VARCHAR	30	Yes	GL account code for debit entry (expense for earnings, liability for deductions)
credit_account	VARCHAR	30	Yes	GL account code for credit entry (liability for earnings, cash/bank for deductions)
cost_center_source	ENUM	-	No	EMPLOYEE_DEFAULT / COMPONENT_SPECIFIC / DIMENSION_SPLIT - How cost center is determined
specific_cost_center	VARCHAR	30	No	Override cost center when cost_center_source = COMPONENT_SPECIFIC
posting_type	ENUM	-	Yes	ACCRUAL / ACTUAL / BOTH - Whether to post accrual entries, actual entries, or both
intercompany_account	VARCHAR	30	No	Intercompany clearing account for cross-entity payroll transactions
effective_from	DATE	-	Yes	Start date of mapping validity
effective_to	DATE	-	No	End date; NULL = currently active

Table 29: GL Account Mapping - Field Definitions

16. Non-Functional Requirements

16.1 Performance

- **NFR-P1:** The payroll engine shall process a payroll run for up to 50,000 employees within 60 minutes, including all component calculations, statutory computations, and report generation.
- **NFR-P2:** Individual payslip generation shall complete within 3 seconds per employee.
- **NFR-P3:** Report generation for standard and statutory reports shall complete within 30 seconds for up to 10,000 employee records.
- **NFR-P4:** The system shall support at least 100 concurrent payroll administrator sessions without performance degradation.
- **NFR-P5:** Exchange rate lookup during payroll calculation shall complete within 50 milliseconds per lookup.

16.2 Security and Data Privacy

- **NFR-S1:** All payroll data (salary amounts, bank details, tax information) shall be encrypted at rest using AES-256 and in transit using TLS 1.3.
- **NFR-S2:** Role-based access control shall enforce data segregation by legal entity, country, and department with field-level permission masking for sensitive salary data.
- **NFR-S3:** All payroll calculation changes and administrative actions shall be recorded in an immutable audit log with user ID, timestamp, before/after values, and IP address.
- **NFR-S4:** Bank account numbers shall be stored in a masked format for display (e.g., XXXXXX1234) with full decryption available only to authorized disbursement roles.
- **NFR-S5:** Multi-factor authentication shall be required for payroll approval and bank file generation operations.

16.3 Availability and Scalability

- **NFR-A1:** The payroll module shall maintain 99.95% uptime, with scheduled maintenance windows limited to non-payroll-processing days.
- **NFR-A2:** The system shall support horizontal scaling of the payroll calculation engine to handle peak month-end processing loads.
- **NFR-A3:** Database failover shall complete within 60 seconds with zero data loss (RPO = 0, RTO = 60 seconds).
- **NFR-A4:** The system shall support multi-region deployment for data residency compliance, with payroll data stored in the country of the legal entity.

16.4 Audit and Traceability

- **NFR-T1:** Every payroll calculation shall produce a computation trace showing the formula, input values, intermediate results, and final amount, stored for a minimum of 7 years.
- **NFR-T2:** Version history shall be maintained for all configuration entities (salary structures, tax slabs, statutory rates, GL mappings) with full before/after change tracking.

- **NFR-T3:** Payroll period data shall be immutable once the period is closed; any corrections must be processed as adjustment transactions in a subsequent period with cross-reference to the original entry.

17. API Specification Overview

The payroll module exposes RESTful APIs following OpenAPI 3.0 specification for integration with adjacent HRMS modules, ERP/accounting systems, banking platforms, and statutory filing portals. All APIs use OAuth 2.0 with JWT authentication, JSON request/response payloads, and support pagination, filtering, and field selection for efficient data retrieval.

API Group	Key Endpoints	Methods	Auth Level
Salary Structure	/api/v1/salary-structures, /api/v1/salary-structures/{id}/components	GET, POST, PUT, PATCH	HR Admin+
CTC Templates	/api/v1/ctc-templates, /api/v1/ctc-templates/{id}/component-mappings	GET, POST, PUT	HR Admin+
Component Master	/api/v1/components, /api/v1/components/{id}, /api/v1/components/{id}/slabs	GET, POST, PUT, PATCH	Payroll Admin+
Payroll Run	/api/v1/payroll-runs, /api/v1/payroll-runs/{id}/calculate, /api/v1/payroll-runs/{id}/approve	POST, PUT, GET	Payroll Admin+
Payroll Inputs	/api/v1/inputs, /api/v1/inputs/bulk-upload, /api/v1/inputs/{id}/approve	GET, POST, PUT	HR / Payroll Admin
Statutory Configuration	/api/v1/statutory/{country}, /api/v1/statutory/{country}/tax-slabs	GET, POST, PUT	Payroll Admin+
Currency & Exchange	/api/v1/currencies, /api/v1/exchange-rates, /api/v1/exchange-rates/latest	GET, POST, PUT	Finance+
Reports	/api/v1/reports/{report_code}, /api/v1/reports/generate, /api/v1/reports/schedule	GET, POST	Payroll Admin+
Compliance	/api/v1/compliance/calendar, /api/v1/compliance/filings, /api/v1/compliance/file	GET, POST, PUT	Payroll Admin+
GL Integration	/api/v1/gl-mappings, /api/v1/journal-entries, /api/v1/journal-entries/post	GET, POST, PUT	Finance+
Employee Payroll	/api/v1/employees/{id}/payslip, /api/v1/employees/{id}/transactions, /api/v1/employees/{id}/ytd	GET	Employee (own), HR+
Dimension Allocation	/api/v1/dimensions, /api/v1/dimensions/{id}/allocations, /api/v1/dimensions/{id}/cost-report	GET, POST, PUT	Finance+

Table 30: API Groups and Key Endpoints